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Paleomagnetism of large igneous provinces: case-study from West Greenland, North Atlantic igneous province

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Abstract

We present new paleomagnetic and multi-model stereo photogrammetry data from lava sequences in the West Greenland part of the North Atlantic igneous province (NAIP). The joint analyses of paleomagnetic and photogrammetric data yield a well-defined paleomagnetic pole located at Lat = 73.6°N, Long = 160.5°E ($N = 44$, $\alpha_{95} = 6.2°$, $K = 13.1$; age ~61–55 Ma), which is statistically indistinguishable from a pole recently obtained for the Eurasian part of the NAIP on Faroe Islands [Riisager et al., Earth Planet. Sci. Lett. 201 (2002) 261–276]. Combining the two datasets we obtain a joint NAIP paleomagnetic pole in Greenland coordinates: Lat = 71.1°N, Long = 161.1°E ($N = 87$, $\alpha_{95} = 4.3°$, $K = 13.6$; age ~61–54 Ma). The results presented here represent the first study in which photogrammetry profiles were photographed at the exact same locations where paleomagnetic fieldwork was carried out, and a direct flow-to-flow comparison of the two datasets is possible. Photogrammetry is shown to be particularly useful because of (i) highly precise dip/strike measurements and (ii) detailed 'field observations' that can be made in the laboratory. Highly precise determination of the structural attitude of well-exposed Kanisut Mb lava sequences demonstrates that their apparently reliable in-field dip/strike measurements typically are up to ~6° wrong. Erroneous dip/strike readings are particularly problematic as they offset paleomagnetic poles without affecting their confidence limits. Perhaps more important for large igneous provinces is the recognition of a variable temporal relationship between consecutive lava flows. We demonstrate how correct interpretation of paleosecular variation, facilitated by the detailed photogrammetry analysis, is crucial for the rapidly emplaced Vaigat Formation lavas. Inaccurate tectonic correction, non-averaged paleosecular variation and unrecognized excursional directions may, perhaps, explain why coeval paleomagnetic poles from large igneous provinces are often discordant. The difference between the joint NAIP paleomagnetitc pole and apparent polar wander path poles suggests that they may be less reliable than suggested by their confidence limits.
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1. Introduction

Large igneous provinces (LIPs) constitute ideal study grounds for paleomagnetism because (i) the

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thick lava sequences provide reliable and detailed records of the paleomagnetic field and (ii) stratification of lavas yields the structural control necessary to detect and correct post-emplacement tectonics. It is, therefore, not surprising that most LIPs have been extensively studied and their paleomagnetic poles dominate global compilations used to discuss apparent and true polar wander paths [1–3] as well as the temporal evolution of the geodynamo [4]. Unfortunately, it is commonly observed that independent coeval paleomagnetic poles from the same LIP are statistically different at the 95% confidence level, indicating that they are less reliable than suggested by their statistical parameters. The fact that paleomagnetic poles from the Parana LIP are statistically distinct has been explained by uncertain tectonic correction [5,6], while the large differences between coeval Deccan traps poles [7] as well as North Atlantic LIP poles [8] have been related to either inadequate demagnetization and/or non-averaged paleosecular variation (PSV).

In this paper we present a new paleomagnetic pole from the North Atlantic igneous province (NAIP) exposed in West Greenland. We pay particular attention to possible error sources related to paleomagnetic pole determination from LIPs, such as incomplete removal of secondary magnetizations, imprecise tectonic corrections, improper handling of PSV and/or unrecognized excursional directions. We show that multi-model stereo photogrammetry provides highly precise measurements of structural attitudes and important data for understanding the temporal relationship of the lava flows and hence the PSV record. Further on, we discuss our new paleomagnetic data in connection with recently compiled apparent polar wander path (APWP) poles of the same age.

2. Geological setting

With an estimated total volume of $\sim 10^7$ km^3 the NAIP (Fig. 1a) represents the third largest magmatic event on Earth during the last 150 Myr [9]. The NAIP formed during two major magmatic phases: a pre-break-up phase ($\sim$62–58 Ma) emplaced prior to North Atlantic break-up and a syn-break-up phase ($\sim$56–54 Ma) contemporaneous with the onset of North Atlantic sea-floor spreading [8,10]. The formation of the NAIP has been linked to the proto-Icelandic hot mantle plume through paleogeographic reconstructions [2,11] and geochemcial observations [12–14].

2.1. West Greenland flood volcanic province

The West Greenland part of the NAIP covers an onshore area of approximately 22 000 km^2 (Fig. 1b; [15]) and a considerably larger offshore area [16]. The Vaigat and Maligat formations represent the pre-break-up phase of volcanism emplaced during magnetochrons C27n and C26r [17–18] and have an estimated stratigraphic thickness of 3–5 km [19]. The earliest volcanism was subaqueous, producing hyaloclastite breccias, while later volcanism was almost entirely subaerial. An unusually large proportion of the pre-break-up volcanism is picritic with a composition indicating an anomalously hot mantle that produced high degrees of partial melting at relatively high pressures [14,20]. Detailed paleomagnetic and photogrammetric studies of the C27n–C26r geomagnetic polarity transition recorded in Vaigat Fm lavas on Nuussuaq Peninsula and Disko Island suggest that the lavas typically erupted in flow fields [21,22], where each flow field represents a series of lavas erupted during a single eruptive event. It is believed that flow fields represent the dominant volcanic emplacement mechanism in continental flood basalt provinces [23]. In our paleomagnetic and photogrammetric analysis of the Vaigat Fm lavas sampled on Svartenhuk we will pay particular attention to the significance of flow fields.

The Kanisut Mb represents syn-break-up volcanism on West Greenland, and ^{40}Ar/^{39}Ar dating of a single comenditic tuff horizon suggests an age of Kanisut Mb of $\sim$55–53 Ma ([24]; A.K. Pedersen, unpublished data). The $\sim$2 km thick sequence of Kanisut Mb lavas is of transitional chemical character [25] and therefore geochemically distinct from the older pre-break-up, more mafic volcanism. The individual Kanisut Mb lava

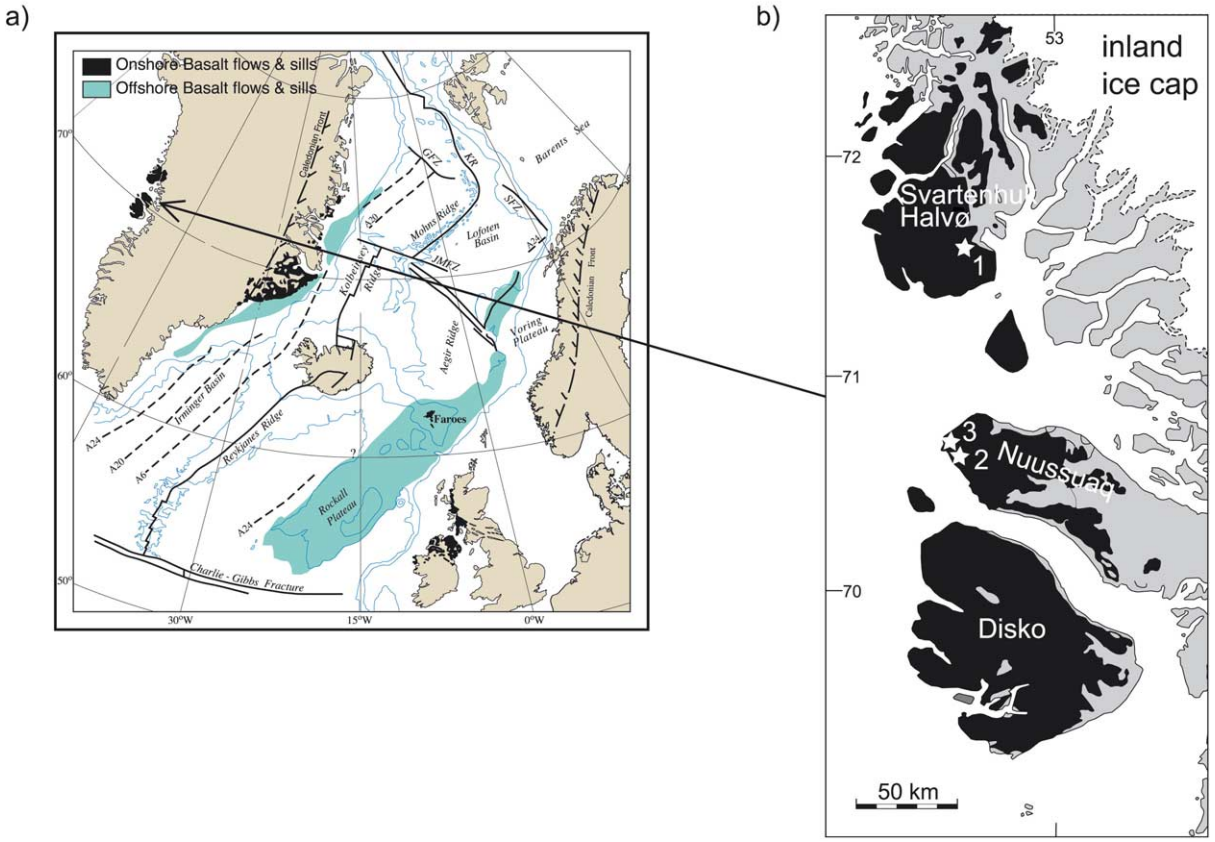


Fig. 1. Maps of (a) the North Atlantic Igneous Province (after [51]); (b) West Greenland flood volcanic province with the geographic position of the sampled profiles shown by stars and numbers (1, Vaigat Fm profiles; 2, Nûluk Mb profile; 3, Kanisut Mb profiles).

flows are thicker and have a significantly different field appearance as compared to the Vaigat Fm picritic lava flows, suggesting a different mode of emplacement.

2.2. Paleomagnetic sampling

For the present study we sampled 59 Vaigat Fm lava flows along two separate profiles in the eastern part of Svartenhuk Peninsula (locality 1 in Fig. 1b), where the lavas are sub-horizontal [26]. The excellent exposure (see Fig. 4) allowed us to sample almost every flow starting from the oldest subaerial lavas. Further south on the Nuussuaq Peninsula (locality 2 in Fig. 1b) five Nûluk Mb lava flows [25] were sampled. Finally, the Kanisut Mb lava sequence was sampled along four short profiles (22 lava flows) at the Nuussuaq Peninsula (locality 3 in Fig. 1b), representing both the youngest and the oldest part of the sequence. The Kanisut Mb flows on the western Nuussuaq Peninsula have a ~15° seaward dip, documenting post-emplacement tectonics (e.g. [27]).

From the altogether 86 lavas, a total of 601 drill cores (five to eight cores per lava flow) were drilled directly in the field with a portable gasoline-powered drill. They were oriented using both magnetic and sun compasses, and the two readings agree well (within ~5°). Due to the close proximity of the geomagnetic pole, and hence relatively small horizontal component of the geomagnetic field at the sampling sites, magnetic compass readings are deemed less reliable and only sun compass orientations are used in our

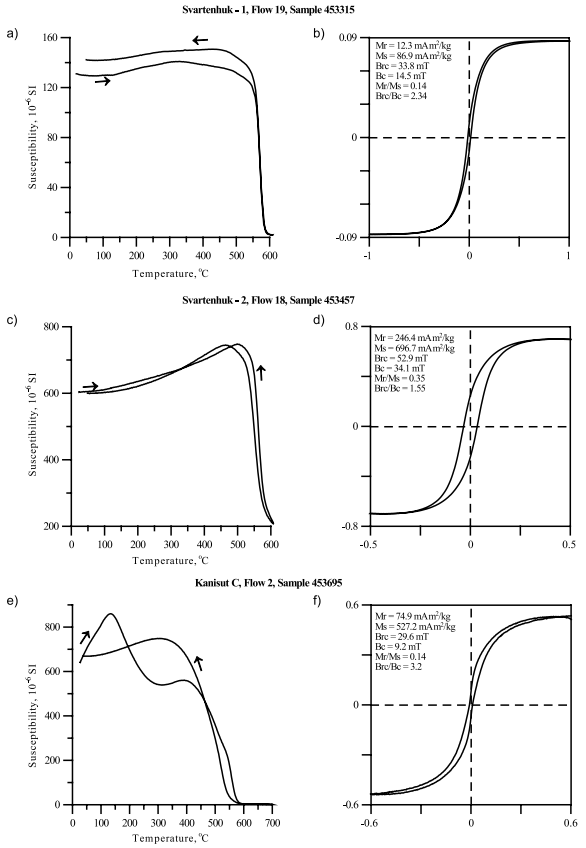


Fig. 2. Typical examples of susceptibility versus temperature curves and hysteresis loops for the (a,b) Svartenhuk-1 profile, (c,d) Svartenhuk-2 profile, (e,f) Kanisut profile. Note, that sample 453457 has close to single-domain hysteresis parameters and Hopkinson peak in $\kappa(T)$ curve indicating single-domain grain size of magnetic carrier, while the sample 453315 $\kappa(T)$ curve, though reversible, has much more muted Hopkinson peak typical for multi-domain particles and hysteresis parameters also indicate close to multi-domain grain size. Sample numbers are GEUS numbers.

analysis. In the laboratory the cores were subsampled into standard cylindrical specimens.

3. Results

3.1. Rock magnetic experiments

Rock magnetic experiments were performed in order to determine the dominant magnetomineralogy of the studied basalts. The experiments include continuous measurements of the temperature dependence of weak-field susceptibility, $\kappa(T)$, in argon using a KLY-2 Kappabridge, and hysteresis measurements performed on a Micromag alternating-gradient force magnetometer.

Vaigat Fm flows are characterized by the presence of a single thermally stable ferromagnetic phase with Curie point of $\sim$570°C, compatible with low-Ti titanomagnetite (Fig. 2a,c). The hysteresis loops have coercivities $B_c \approx 10$–30 mT (Fig. 2b,d) and ratios of saturation remanent magnetization over saturation magnetization between 0.1 and 0.4, indicating a mixture of single-domain, pseudo-single-domain and multi-domain grain sizes [28]. These observations are in good accordance with direct scanning electron microscopy results [29], indicating that the magnetomineralogy of Vaigat Fm lavas is dominated by oxy-exsolved titanomagnetites.

The $\kappa(T)$ curves for Kanisut Mb and Nûluk Mb lava flows are more diverse than the Vaigat Fm curves, indicating both primary titanomagnetite and less titanium-rich titanomagnetite phases (Fig. 2e). In addition, some Kanisut Mb lavas may contain maghemite, evidenced by irreversible $\kappa(T)$ curves. The hysteresis loops of the Kanisut Mb and Nûluk Mb flows range from simple symmetrical loops to slightly wasp-waisted (Fig. 2f).

3.2. Demagnetization procedures

Two samples per lava flow were subjected to thermal cleaning in air in a laboratory-built furnace at the paleomagnetic laboratory at the University of California, Santa Cruz, with remanence measurements performed in a magnetically shielded room on a JR5 spinner magnetometer. All remaining samples were subjected to three-axis alternating-field (AF) demagnetization and measured with 2G cryogenic magnetometers at the University of Lund, Sweden, and University of California, Santa Cruz. Demagnetization results obtained in different laboratories and with different methods were found to be in excellent agreement. The direction of the characteristic remanent magnetization (ChRM) was calculated using standard principal-component analysis [30]. All accepted ChRM directions are based on at

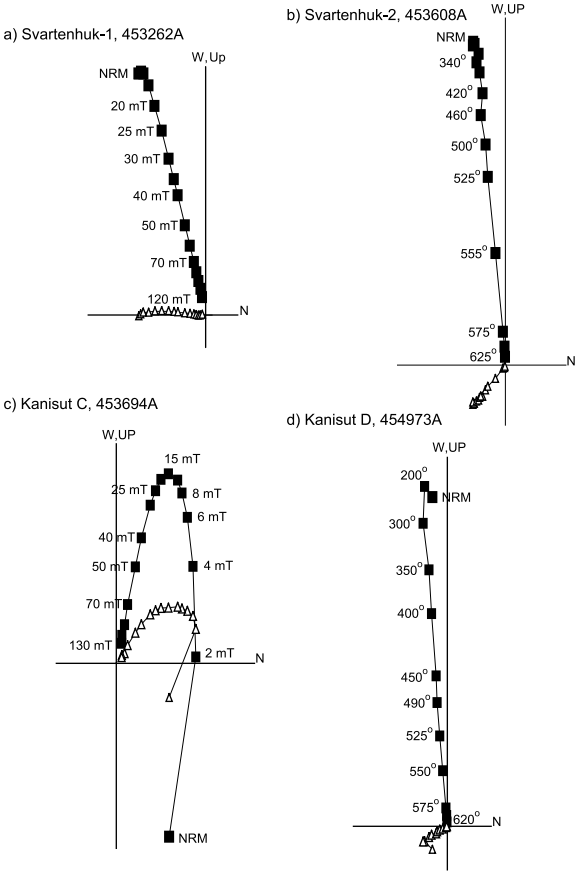


Fig. 3. Examples of orthogonal projections of alternating-field and thermal demagnetizations. Solid (open) symbols on orthogonal projection correspond to vertical (horizontal) plane. Sample numbers are GEUS numbers.

least five demagnetization steps and have maximum angle of deviation less than 3°.

The demagnetization results closely follow rock magnetic observations. Vaigat Fm lava flows have essentially univectorial demagnetizations (Fig. 3a,b) with median destructive fields in the range 30–50 mT and unblocking temperatures, 400–570°C, in accordance with Ti-poor titanomagnetite being the main remanence carrier. Flow-mean directions are well defined, as evidenced by their 95% confidence cones (α_{95}) that on average are only 2.3° (see Table 1A). The Nûluk Mb and Kanisut Mb basalts display more variable behavior from ideal one-component natural remanent magnetization to large secondary overprints, most probably of viscous origin (Fig. 3c,d). The secondary overprint could not be entirely removed for ~5% of the Kanisut Mb samples and these samples are, therefore, not included in the following analysis (Table 1C).

3.3. *Multi-model stereo photogrammetry observations*

Multi-model stereo photogrammetry has previously been compiled along several 5–80 km long sections through the West Greenland flood volcanic province, allowing detailed mapping of lateral changes in facies, lithology and tectonics and giving a unique insight into the highly dynamic state of the province during the volcanic emplacement [21,31,32]. The results presented here represent the first study in which photogrammetry profiles were photographed at the exact same location where paleomagnetic fieldwork was carried out, and a direct flow-to-flow comparison of the two datasets is possible.

The profiles were photographed from a helicopter with stereoscopic overlap and in color; photographs were later set up for the multi-model photogrammetry analyses at the Technical University of Denmark. The analyses allow the exact position of any point on the profile to be determined in three-dimensional space with a relative precision better than ~0.5 m. By digitizing ~100 points along the best-defined and most easily traced flow boundaries over their entire length (typically ~1 km), it is possible to define the attitude of strata with a precision better than ~0.5°. For the Svartenhuk-1 profile the dip/strike for three lavas was found to be 8.7°/194°, 6.7°/216°, and 7.8°/212°, respectively. The ~2° variation in dip reflects slight changes in the paleo-topography related to small-scale basin subsidence and the piling up of lavas as the sequence was emplaced. The post-emplacement tectonic rotation of the lava pile is best estimated from the mean dip/strike of 7.6 ± 2.9°/206 ± 22° (uncertainties are ±2σ). At the Svartenhuk-2 profile the dip/strike was defined for four lavas (see Fig. 4) and their mean is 1.5 ± 1.0°/280 ± 37.8°. All photogrammetry dip/strike measurements for the profiles of the Nûluk Mb and Kanisut Mb are shown in Fig. 5 together with 'in-field' dip measurements

Table 1A
Paleodirectional data for the Vaigat Fm lava flows

DG	Flow #	Alt. (m)	*N/n*	I_{geo} (°)	D_{geo} (°)	I_{str} (°)	D_{str} (°)	κ	α_{95} (°)	VGP Lat (°)	VGP Long (°)
Svartenhuk-1											
DG1	1	420.9	10/10	−71.8	182.5	−78.4	167.5	285.8	2.9	84.2	180.4
	2	429.8	0/6			unstable					
DG2	3	435.1	3/3	−78.5	203.8	−86.1	198.8	1229.8	3.5	78.6	318.7
	4	439.3	6/7	−78.9	197.7	−86.3	179.9	2065.5	1.5	78.9	306.0
	5	446.8	4/5	−77.4	201.4	−84.9	193.9	1177.2	2.7	80.9	321.4
	6	450.3	6/6	−77.7	193.5	−84.8	174.5	860.5	2.3	81.6	299.3
	7	455.2	7/7	−77.8	187.9	−84.5	162.2	554.8	2.6	81.3	283.6
Mean DG2			26	−78.1	195.3	−85.3	177.1	838.1	0.9	80.8	303.1
DG3	8	460.4	5/5	−74.4	184.4	−81.0	166.2	1438.3	2.0	85.6	235.3
	9	468.3	2/4	−70.6	198.7	−78.1	193.9			83.5	70.6
	10	471.9	5/5	−75.1	191.8	−82.2	177.7	1127.3	2.3	86.7	295.7
	11	473.9	5/6	−76.7	191.6	−83.8	173.7	268.0	4.7	83.6	293.9
	12	480.3	6/6	−73.7	190.2	−80.8	177.2	310.3	3.8	89.0	248.1
	13	490.5	6/6	−73.4	183.3	−80.0	166.4	1617.8	1.7	85.5	210.3
	14	502.2	8/8	−75.0	179.3	−81.1	156.8	162.5	4.4	82.8	235.9
	15	504.3	8/8	−73.6	185.6	−80.3	169.7	991.5	1.8	86.7	216.4
Mean DG3			45	−74.6	188.4	−81.5	172.6	210.2	1.5	87.1	259.2
DG4	16	543.3	6/6	−63.1	149.2	−66.4	134.5	357.6	3.5	59.4	193.1
	17	594.6	5/5	−60.7	145.1	−63.6	131.7	1002.4	2.4	55.2	193.4
Mean DG4			11	−62.0	147.3	−65.1	133.2	455.2	2.1	57.4	193.1
DG5	18	634.2	4/5	−53.9	131.2	−55.2	120.6	447.1	4.4	43.2	199.6
	19	658.1	5/5	−60.4	130.8	−61.4	117.2	163.7	6	48.4	207.0
	20	690.3	6/6	−57.5	129.9	−58.5	117.8	752.6	2.4	45.6	204.4
Mean DG5				−57.5	130.6	−58.6	118.5	266.7	2.3	45.9	203.6
Svartenhuk-2											
DG1	1	652.5	6/6	−78.4	180.6	−78.0	173.5	3747.7	1.1	84.8	155.1
	2	657.8	9/9	−79.7	180.5	−79.3	172.5	1535.1	1.3	86.6	176.2
Mean DG1			15	−79.2	180.5	−78.8	172.9	1886.5	0.9	85.9	165.5
DG2	3	660.3	8/8	−62.4	270.5	−63.9	270.0	714.5	2.1	42.7	18.0
	4	664.4	5/5	−61.7	269.2	−63.2	268.6	1410.4	2.0	42.3	19.6
	5	666.0	5/5	−61.2	274.4	−62.7	274.1	707.1	2.9	40.2	15.2
Mean DG2*			18	−61.9	271.2	−63.4	270.7	786.9	1.2	41.9	17.7
DG3*	6	669.6	8/8	−59.1	267.4	−60.6	266.8	1211.6	1.6	40.0	23.0
DG4*	7	678.9	9/9	−71.1	286.2	−72.6	286.7	355.5	2.7	49.2	356.8
DG5	8	683.9	7/7	−66.9	286.1	−68.4	286.5	3230.4	1.1	43.6	0.9
	9	691.5	8/8	−67.2	292.1	−68.7	292.9	973.6	1.8	42.3	355.7
	10	697.9	5/5	−67.1	288.6	−68.6	289.1	855.9	2.6	43.2	358.7
	11	699.4	6/6	−68.2	286.8	−69.7	287.3	571.4	2.8	45.1	359.2
	12	701.1	11/12	−65.7	285.8	−67.2	286.1	467.1	2.1	42.1	2.1
	13	702.9	8/8	−66.9	283.7	−68.4	283.9	823.9	1.9	44.3	2.9
	14	705.9	7/7	−68.9	283.7	−70.4	283.9	1278.1	1.7	46.9	1.1
	15	709.0	7/7	−68.2	285.1	−69.7	285.4	1303.8	1.7	45.6	0.6
	16	715.7	8/8	−68.4	283.9	−69.9	284.1	870.1	1.9	46.2	1.4
	17	721.9	9/9	−68.6	287.2	−70.1	287.7	923.6	1.7	45.5	358.5
	18	724.4	8/8	−69.0	280.7	−70.5	280.7	514.6	2.4	47.9	3.4
	19	729.0	8/8	−69.6	286.1	−71.1	286.5	225.9	3.7	47.2	358.4
	20	736.7	6/7	−70.0	284.7	−71.5	285.0	1256.8	1.9	48.1	359.1
	21	741.9	7/7	−67.6	289.2	−69.1	289.8	1041.4	1.9	43.6	357.8
	22	746.3	8/8	−69.2	290.2	−70.7	290.9	6387.1	0.7	45.5	355.6
	23	750.9	7/7	−69.9	286.4	−71.4	286.9	2019.1	1.3	47.5	357.9

Table 1A (*Continued*).

DG	Flow #	Alt. (m)	*N/n*	I_{geo} (°)	D_{geo} (°)	I_{str} (°)	D_{str} (°)	κ	α_{95} (°)	VGP Lat (°)	VGP Long (°)
Mean DG5*			120	−68.2	286.4	−69.7	286.8	643.0	0.5	45.2	359.6
DG6	24	759.2	8/8	−68.2	279	−69.7	278.9	1006.5	1.7	47.3	5.6
	25	762.2	9/9	−67.7	281.3	−69.2	281.4	517.9	2.3	46.0	4.2
	26	763.4	8/8	−67.5	282.7	−69.0	282.9	1007.5	1.7	45.3	3.2
	27	766.1	9/9	−67.1	280.8	−68.6	280.8	2356.7	1.1	45.3	5.2
	28	768.6	9/9	−69.6	277.8	−71.1	277.6	645.6	2.0	49.4	4.9
	29	769.3	7/7	−66.9	276.9	−68.4	276.7	2553.8	1.2	46.2	8.7
	30	773.0	8/8	−67.5	275.1	−69.0	274.7	856.3	1.9	47.5	9.6
Mean DG6*			58	−67.8	279.2	−69.3	279.1	803.5	0.7	46.7	5.9
DG7*	31	778.5	7/7	−67.8	283.5	−69.3	283.7	1481.8	1.6	45.5	2.3
DG8*	32	779.6	9/9	−69.2	275.6	−70.7	275.2	1057.7	1.6	49.6	7.4
DG9*	33	796.4	6/6	−72.3	270.1	−73.8	269.2	656.4	2.6	55.4	8.0
DG10	34	806.5	7/7	−84.8	148.6	−83.7	138.3	497.6	2.7	77.9	262.5
	35	816.1	6/6	−84.5	159.8	−83.6	148.0	1830.4	1.6	80.0	263.8
			13	−84.7	153.9	−83.6	142.8	781.8	1.5	78.9	262.5
DG11	36	828.1	7/7	−84.3	124.3	−82.8	119.5	1234.4	1.7	73.4	257.2
DG12	37	837.1	8/8	−81.2	125.9	−79.8	122.2	3534.5	0.9	71.8	239.1
	38	841.5	7/7	−84.2	125.8	−82.8	120.6	370.3	3.1	73.7	257.0
Mean DG12			15	−82.6	125.9	−81.2	121.7	609.3	1.6	72.9	246.9
DG13	39	856.9	10/10	−73.8	142.5	−72.6	139.2	432.9	2.3	68.6	197.3
Kuugannguup Innartaa											
KI 1	1	230	4/4	43.1	330.6	43.1	330.6	149.7	7.5	41.7	161.7
KI 2	2–5	250	16/16	48.9	337.5	48.9	337.5	262.1	2.3	47.6	154.7
KI 3	6	290	4/4	43.6	8.5	43.6	8.5	29.6	17.2	44.9	114.3
KI 4	7	305	5/5	63.1	347.5	63.1	347.5	441.2	3.7	63.6	145.3
KI 5	8	330	4/4	52.9	348.0	52.9	348.0	453.0	4.3	52.6	141.6
KI 6	9–12	338	21/21	65.7	10.8	65.7	10.8	416.4	1.5	67.1	106.3
Nuusap Qaqqarsua											
NQ 2	4–6	865	15/15	70.9	4.9	70.9	4.9	145.0	3.2	74.7	114.5
NQ 3	7	890	5/6	59.9	8.1	59.9	8.1	179.4	5.7	60.1	112.8
NQ 4	8–9	905	6/7	67.0	20.9	67.0	20.9	561.1	2.8	67.1	88.8
NQ 5	10	950	4/4	71.7	6.2	71.7	6.2	1008.0	2.9	75.8	111.0
NQ 23	42	1150	4/4	−72.4	153.2	−72.4	153.2	632.4	3.7	72.8	180.4
NQ 24	43	1160	3/3	−76.3	132.4	−76.3	132.4	61.6	15.8	71.1	219.1
NQ 25	44	1170	4/4	−65.9	156.1	−65.9	156.1	167.4	7.1	65.0	164.9
NQ 26	45	1178	4/4	−57.7	151.9	−57.7	151.9	484.3	4.2	54.7	164.9

DG is a directional group; Alt is the altitude of flow in meters measured in multi-model stereo photogrammetry; *N/n* is the number of treated/used samples; I_{geo}, D_{geo} are in situ inclination and declination of flow-mean ChRMs; I_{str}, D_{str} are inclination and declination of the tectonically corrected flow-mean ChRMs; κ, α_{95} are the precision parameter and the confidence cone of the Fisher statistics [36]; VGP Lat, VGP Long are the latitude and longitude of the south VGP. The geographic coordinates are: 71.47°N, 306.05°E for Svartenhuk-1; 71.74°N, 305.70°E for Svartenhuk-2. Tilt corrections are: for Svartenhuk-1, dip = 7.62° with azimuth 206.4°; for Svartenhuk-2, dip = 1.53° with azimuth 280.5°. Flows marked with asterisk (*) yield excursional directions and were excluded from further analyses. Lava flows of both Svartenhuk profiles are the subject of this study, while additional 14 cooling units of Kuugannguup Innartaa and Nuusap Qaqqarsua are taken from [22].

taken from the geological map. A comparison of photogrammetry and in-field dip/strike measurements (Fig. 5b) suggests that although the in-field readings are taken at the most suitable and well-exposed locations, they are at times off by 5–6°. We believe this difference is caused by in-field dip/strike measurements being limited to the two-dimensional exposure and short distances over which the measurements are made. The precise photogrammetry readings not only allow us to define tectonic corrections more accurately but also show that the entire Kanisut Mb sampling

area has tilted as one coherent block, i.e. having the same tectonic dip/strike throughout (see Fig. 5), which is not immediately apparent from the much more scattered in-field dip/strike measurements. Finally, we note that although the dip/strike of Nûluk Mb is precisely defined (Fig. 5), it is offset compared to Kanisut Mb lavas (Fig. 5), indicating that the sampling area has suffered a land-slide during the Holocene, which may have included vertical-axis rotations. The paleomagnetic data from Nûluk Mb flows are, therefore, not included into PSV analysis and paleomagnetic pole determination but considered for magnetostratigraphy purposes only.

Another major advantage of photogrammetry is the possibility to make 'field observations' in the laboratory, which are not always viable during fieldwork. For example, in photogrammetry analyses it is possible to 'zoom out' and obtain an overview (similar to Fig. 4) that allows detection of large-scale geological features, for example faults, which are not always easily recognized while working on the profiles. 'Zooming in' on finer details, it is possible to discover or re-discover details such as soil horizons, variations in lava thickness, color and morphology, which may have gone unnoticed during fieldwork. Such 'field observations' in the laboratory are of particular importance when paleomagnetic analyses have been completed. As shown below, comparisons between the photogrammetry and paleomagnetic datasets help a great deal in understanding the PSV record in the rapidly emplaced Vaigat Fm lavas.

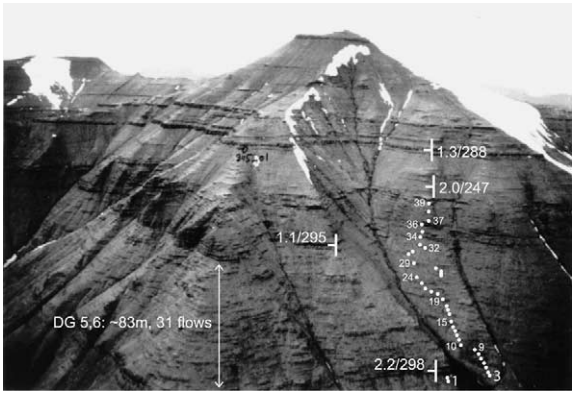


Fig. 4. Photograph of the Svartenhuk-2 profile taken from a helicopter for multi-model stereo photogrammetry. Dips/strikes measured in photogrammetry are shown. Also shown are the directional groups 5 and 6 (DG5,6). Note the remarkable correspondence between DG boundary (vertical arrow) and change in color of the volcanic sequence. White dots depict position of each flow in the paleomagnetic profile.

3.4. Defining independent paleomagnetic field directions

The flow-mean directions corrected for tectonic dip/strike (i.e. in stratigraphic coordinates) and their corresponding statistical parameters are summarized in Table 1A and Fig. 6a,b for the two Svartenhuk profiles, in Table 1B for Nûluk Mb and in Table 1C and Fig. 6c for the four Kanisut Mb profiles. An instantly recognizable feature of the Vaigat Fm profiles (Fig. 6a,b) is that consecutive flows often have similar flow-mean directions. This feature is commonly ob-

Table 1B
Paleodirectional data for lava flows of Nuluk Mb

Flow #	Alt. (m)	*N*/*n*	I_{geo} (°)	D_{geo} (°)	I_{str} (°)	D_{str} (°)	κ	α_{95} (°)	VGP Lat (°)	VGP Long (°)
Nuluk										
1	31.8	3/3	−61.4	257.2	−79.7	220.0	253.5	7.8	76.8	19.0
2	130.7	4/4	−75.8	230.3	−75.0	136.8	358.6	4.9	71.0	209.7
3	137.2	3/3	−65.1	216.8	−67.7	165.6	238.1	8.0	68.9	151.5
4	160.0	3/3	−70.6	262.2	−85.1	153.9	148.7	10.1	78.6	283.3
5	174.2	3/3	−68.4	230.7	−74.1	164.6	1079.2	3.8	77.9	164.5

Notation is the same as for Table 1A. The geographic coordinates of Nuluk Mb lava flows are 70.64°N, 305.50°E. Tilt correction is dip = 21.5° with azimuth 275.0°. The Nuluk Mb sampling area has suffered a land-slide and the paleomagnetic data are, therefore, not included in the PSV analysis and paleomagnetic pole determination but considered for magnetostratigraphy purposes only.

Table 1C
Paleodirectional data for the four Kanisut Mb profiles

Flow #	Alt. (m)	N/n	I_{geo} (°)	D_{geo} (°)	I_{str} (°)	D_{str} (°)	κ	α_{95} (°)	VGP Lat (°)	VGP Long (°)
Kanisut A										
1	10	7/8	−70.2	233.6	−67.0	196.3	312.6	3.4	67.6	96.9
2	10	8/8	−66.2	151.4	−51.9	146.4	92.9	5.8	47.7	169.4
3	10	7/7	−73.2	151.7	−58.9	145.2	162.9	4.7	54.2	174.2
4	10	8/8	−78.3	222.8	−70.7	174.7	204.8	3.9	74.1	136.5
Kanisut B										
1	0	6/6	−78.3	111.3	−64.6	125.2	411.4	3.3	54.6	201.9
2	10	7/7	−67.9	202.6	−59.2	179.0	308.4	3.4	59.3	127.0
Kanisut C										
1	130	7/7	−76.3	211.7	−66.9	172.9	419.8	2.9	68.7	138.2
2	120	8/8	−71.7	310.7	−86.5	281.4	152.4	4.5	68.2	324.2
3	110	7/7	−73.9	314.5	−88.9	271.3	520.4	2.7	70.5	311.9
4*	90	7/7	−70.3	43.2	−66.1	81.2	727.9	2.2	42.3	243.1
5	82	7/7	−82.8	32.1	−74.9	109.6	238.8	3.9	62.0	232.9
6	70	6/7	−72.6	260.2	−74.4	206.4	697.9	2.5	75.6	64.9
7	50	7/7	−67.9	241.4	−66.6	204.0	2119.1	1.3	65.7	85.1
8	40	6/6	−65.8	239.4	−64.5	205.7	1196.3	1.9	62.7	84.7
9	30	6/10	−79.8	67.5	−68.9	109.8	26.8	13.2	54.6	222.1
Kanisut D										
1	10	7/7	−73.4	165.4	−60.4	148.8	97.9	6.1	56.8	170.6
2*	10	7/7	−13.7	110.7	0.2	111.2	76.1	6.9	−6.8	15.3
3	10	6/6	−83.9	231.1	−75.5	152.9	427.5	3.2	76.8	191.7
4	10	5/5	−76.5	145.4	−62.2	136.9	1478.5	1.9	55.6	186.8
5	10	5/5	−54.9	204.4	−49.1	187.0	96.8	7.8	49.2	116.2
6	10	6/6	−78.7	164.5	−65.3	144.6	293.9	3.9	61.4	180.4
7	10	8/8	−73.1	267.8	−78.9	210.0	274.2	3.4	79.5	31.9

Notation is the same as for Table 1A. The geographic coordinates are: 70.71°N, 305.44°E for Kanisut A; 70.68°N, 305.43°E for Kanisut B; 70.67°N, 305.44°E for Kanisut C; 70.64°N, 305.50°E for Kanisut D. Tilt corrections are: for Kanisut A, dip = 14.6° with azimuth 317.0°; for Kanisut B, dip = 14.4° with azimuth 317.0°; for Kanisut C, dip = 15.3° with azimuth 317.3°;for Kanisut D, dip = 14.6° with azimuth 308.5°.

served in LIP lava sequences and often taken to indicate that the series of flows were erupted in a short period of time and they essentially record the same paleomagnetic field [8,22,33]. In our study this interpretation is also supported by photogrammetry observations. The most remarkable example is the ~83 m thick series of 31 lavas of the Svartenhuk-2 profile recording largely similar directions (directional groups (DGs) 5 and 6 in Fig. 6b), and having a grayish color distinctly lighter than flows above and below (Fig. 4). We take the joint paleomagnetic and photogrammetry observations as strong evidence that flows forming DGs 5 and 6 stem from a single eruptive event, i.e. forming a single flow field. The fact that the sequence records a slight angular change of virtual geomagnetic poles (VGPs) of ~5° suggests that the flow field was emplaced over a long enough period for ~5° of secular variation to be recorded. Assuming that secular variation rate during the mid-Paleocene was similar to that of the Holocene (5°/100 yr [33]), it probably took less than ~100 yr to accumulate the DGs 5 and 6 flow field (Fig. 4). Further details regarding flow fields of the Vaigat Fm, including geochemical and paleo-intensity variations, will be the subject of a separate paper.

For the following analysis, DGs are defined as consecutive flows whose paleomagnetic directions are statistically indistinct at the 95% confidence level as determined by the Watson test [34]. In the Watson test the F statistic is calculated based on R values and number of samples of each consecutive flow and compared to the tabulated F

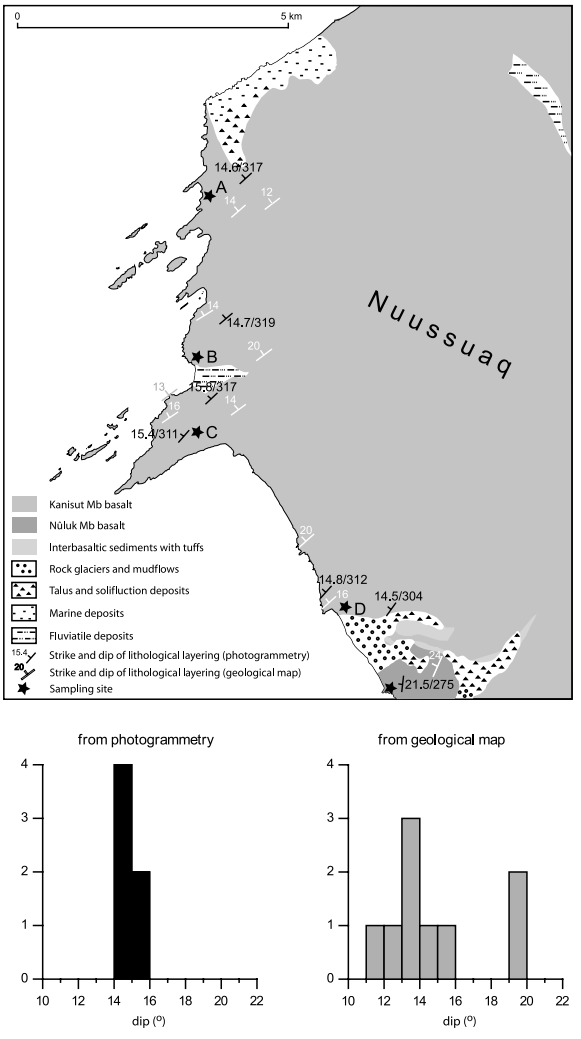


Fig. 5. (a) The western Nuussuaq coastal map. Dips/strikes measured in photogrammetry are shown in black; in-field measurements are shown in gray. (b) Histograms of in-field and photogrammetry dip measurements.

value for the given degree of freedom. To verify the outcome of the Watson test, a non-Fisherian bootstrap test [35] was also performed and yielded the exact same DGs. From the total of 59 sampled flows of the two Svartenhuk profiles, we obtain 17 DGs (Table 1A and Fig. 6a,b). For Kanisut Mb (Fig. 6c) and Nûluk Mb no consecutive lava flows record similar directions, indicating that these were emplaced in a distinctly different volcanological setting, with much slower eruption rates than Vaigat Fm.

3.5. Paleosecular variation

For the further analysis we combine the Vaigat Fm DGs and Kanisut Mb lava flows of this study with 14 Vaigat Fm DGs of stable polarity recently obtained from Nuussuaq Peninsula and Disko Island (Fig. 1b; [22]). A large paleomagnetic dataset of 54 independent readings of the paleomagnetic field is then obtained, site-mean directions and corresponding VGPs of which are shown in Fig. 7 and listed in Table 1A,C. In order to detect anomalous VGPs due to excursional flow-mean directions, we used a cut-off angle of 40°. We find that eight DGs of the Svartenhuk-2 profile (DGs 2–9; Table 1A), and two Kanisut Mb flows (flow 4 of Kanisut C and flow 2 of Kanisut D; Table 1C) lie outside of the cut-off angle (Fig. 7b) and most likely record excursional directions. These DGs/flows were excluded from the further analyses and only 44 data points remain in the dataset.

The PSV is usually expressed by the angular standard deviation (ASD) of the VGP distribution with the implicit assumption that the distribution is Fisherian [36]. In order to test whether our dataset is Fisher-distributed we produce quantile–quantile plots for exponential co-latitude and uniform longitude (Fig. 8). The M_u and M_e values lower than threshold values of 1.207 and 1.094 respectively indicate that a Fisher distribution hypothesis cannot be rejected at 95% confidence level [35]. The ASD value is 22.6° with a 95% confidence limit of +3.9°/−2.9° [37]. This value is in excellent agreement with estimates derived from the Eurasian part of the NAIP on Faroe Islands 21.7°+3.9°/−2.8° [8], and it also coincides well with the PSV estimate of 21.1° predicted by the G-model of [38] for this paleolatitude (53.9°).

3.6. Magnetostratigraphy

A major objective of sampling the oldest subaerial Vaigat Fm lavas on Svartenhuk Peninsula was to sample the C27n–C26r geomagnetic polarity reversal, found in Vaigat Fm lavas at Nuussuaq and Disko [17,22]. However, both Svartenhuk profiles were found to be entirely of reverse

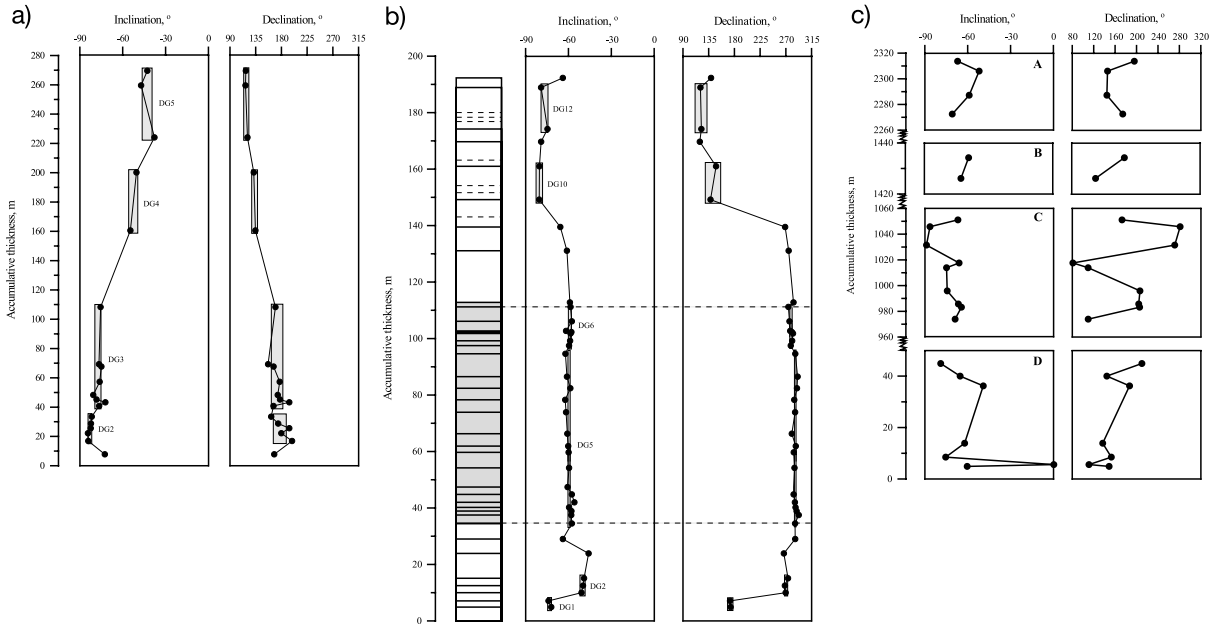


Fig. 6. Direction of characteristic remanent magnetization in (a) Svartenhuk-1; (b) Svartenhuk-2; (c) four Kanisut profiles. Rectangular boxes indicate DGs. Map thickness calculated using dip/strike measurements and *x*,*y*,*z* coordinates from photogrammetry analyses. Dashed lines are lava flows that were not sampled for paleomagnetic study, but whose thickness was measured in photogrammetry analyses.

polarity, which indicates commencement of volcanic activity on Svartenhuk Peninsula in magnetochron C26r and, hence, postdates the initiation of volcanism on Nuussuaq Peninsula and Disko Island. The Nûluk Mb flows, representing the youngest Paleocene lavas of Maligat Fm, are also reversely magnetized, suggesting that pre-break-up volcanism on West Greenland ended in C26r (see also [18]). This is dissimilar to pre-break-up NAIP volcanism on Faroe Islands, which continued through C25n [8]. Our paleomagnetic data from Kanisut Mb (Fig. 6c) represent the first comprehensive magnetostratigraphic coverage of West Greenland syn-break-up volcanism. The reverse polarity of Kanisut Mb is in good accordance with two $^{40}Ar/^{39}Ar$ ages ([24]; A.K. Pedersen, personal communications) derived from the central part of the sequence, suggesting that the entire Kanisut Mb was erupted during magnetochron C24r. This places the onset of syn-break-up volcanism on West Greenland in good temporal relation with the syn-break-up volcanism on the Faroe Islands [8] and East Greenland [39] as well as with the timing of the North Atlantic break-up, where the oldest ocean-floor magnetic lineation is C24r [40].

The eight consecutive DGs recording excursional directions in the Svartenhuk-2 profile most likely represent one of the seven cryptochrons in the magnetochron C26r, and the two excursional Kanisut Mb flows two out of the 11 cryptochrons in C24r [41]. Unfortunately, exact correlation of excursions to the geomagnetic polarity time scale is not possible at this stage.

3.7. Paleomagnetic pole

To obtain a paleomagnetic pole we must transfer the paleomagnetic data from geographic into stratigraphic coordinates. The tectonic correction used in this study is derived from precise photogrammetry observations and based on three to four estimates at each sampling profile that allow us to average out variable local-scale paleo-topography as the lava sequences were piling up. A regional-scale paleo-topography, which is difficult

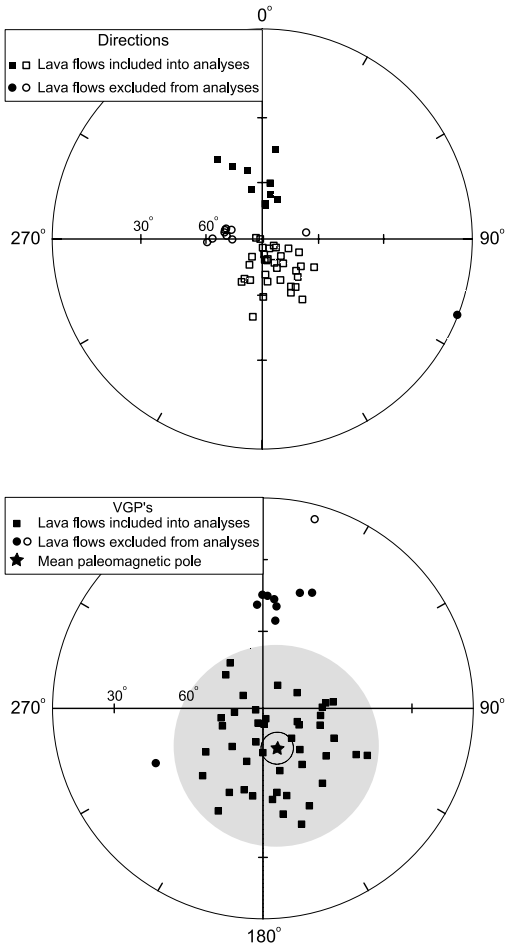


Fig. 7. (a) Stereo diagram of mean directions for the combined dataset. Open (solid) symbols correspond to negative (positive) inclination. (b) Equal-area diagram of corresponding VGPs and the 40° cut-off angle. The mean paleomagnetic pole (star) and its 95% confidence limit are also given.

to estimate for, for example, stratoid basalts [42], is unlikely to exist in West Greenland flood volcanic province given its size ($>22\,000$ km^2). In order to verify the tectonic correction, the fold test [43] was applied to all 44 independent and non-excursional cooling units and found to be positive with the maximum untilting at 99.4% (Fig. 9). The tectonically corrected mean VGP of Kanisut Mb and mean VGP of Vaigat Fm are statistically indistinguishable, as seen from Fig. 10a (poles 3 and 4 in Table 2). Resulting mean direction in stratigraphic coordinates is

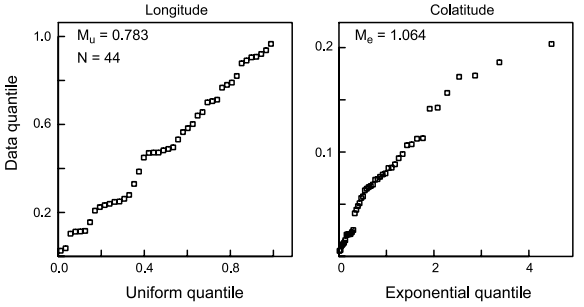


Fig. 8. Quantile–quantile plots assessing goodness-of-fit of the Fisher distribution to the West Greenland VGPs. M_u is lower than 1.207 and M_e is lower than 1.094, hence hypothesis of Fisher distribution cannot be rejected.

Inc = −70.0°, Dec = 163.1° ($N = 44$, $\alpha_{95} = 4.1°$, $K = 29.1$). The corresponding paleomagnetic pole has coordinates: Lat = 73.6°, Long = 160.5° ($N = 44$, $B = 383$, $\alpha_{95} = 6.2°$, $K = 13.1$). The time span covered by studied flows as well as PSV analyses demonstrates that this pole represents a time-averaged field. The significance of the new West Greenland paleomagnetic pole and its correspondence with other NAIP and APWP poles of the same age is discussed below.

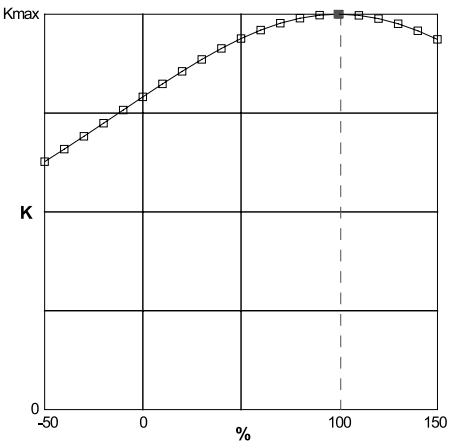


Fig. 9. Fold test of [43] on 44 independent and non-excursional cooling units used to calculate PSV and paleomagnetic pole (Table 1A,C). The test was found positive with the maximum untilting at 99.4% per cent.

4. Discussion

The first compilations of paleomagnetic poles, published in the 1950s [44,45], played an important role in the development of plate tectonics, giving quantitative evidence for continental drift. Since then the continual accumulation of paleomagnetic poles has led to a significant improvement in both temporal and spatial resolution of the APWPs for almost all major lithospheric plates [2,3]. In fact, recent APWPs quote confidence limits as low as 2–4°, which potentially open new areas of research, including highly precise paleogeographic reconstructions of inter-continental deformation and rifting at continental margins. It is, however, important to recognize that many existing paleomagnetic poles may bear large uncertainties, which are not always reflected in their confidence limits. It has, for example, been suggested that sedimentary paleomagnetic poles may be systematically biased [46] and that geomagnetic and geologic inferences based on APWPs, which include these data, could be erroneous [47,48]. Unlike sediments, intrusive magmatic rocks reliably record the paleomagnetic field, but, on the other hand, lack tectonic control and it is, therefore, important to also treat these data with care [49]. Lava sequences represent the ideal case where both structural control and reliable records of the paleomagnetic field are available and paleomagnetic poles from lavas are, therefore, often argued to be the most reliable source of data [1,48]. It is, hence, alarming that independent paleomagnetic poles from different lava sequences within the same LIP are often found to be statistically distinct [6–8].

Based on the West Greenland data we will briefly discuss possible error sources which can potentially cause LIP paleomagnetic poles to be offset. First of all, it has been suggested that flow fields are common in LIPs [22] and it is, therefore, important to consider the complexities that flow fields add to the temporal relationship between consecutive lavas in paleodirectional studies. If we had followed the procedures used in most previously published LIP studies and treated each Vaigat Fm lava as an independent reading of the paleomagnetic field, we would not only have underestimated the PSV but also obtained an erroneous paleomagnetic pole dominated by the paleomagnetic field directions recorded in massively thick flow fields. Although grouping of lavas into DGs improves this problem, often not even the DGs are temporally independent as (i) different flow fields can be closely spaced in time, and (ii) a single flow field can contain more than one DG

Table 2
Summary of the NAIP paleomagnetic poles

Paleomagnetic pole	Age (Ma)	N	Lat (°N)	Long (°E)	α_{95} (°)	κ	ASD (°)	Reference
1. Vaigat Fm (Svartenhuk)	60–61	10	76.2	217.9	9.8	25.1	16.3+7.7/−4.0	This study
2. Vaigat Fm (Nuussuaq and Disko)	60–61	14	64.8	141.5	9.2	19.8	18.3+6.7/−3.9	[22]
3. Combined Vaigat Fm (1+2)	60–61	24	72.9	161.2	8.0	14.7	21.3+5.4/−3.6	This study
4. Kanisut Mb	~55	20	74.6	159.4	10.3	11.1	24.6+7.1/−4.5	This study
5. West Greenland (3+4)	55–61	44	73.6	160.5	6.2	13.1	22.6+4.0/−2.9	This study
6. Faroe Islands*	55–58	43	68.1	162.0	6.0	14.3	21.7+3.9/−2.8	[8]
7. Joint-NAIP (5+6)	55–61	87	70.9	161.3	4.3	13.6	22.0+2.5/−2.1	This study
8. APWP-I**	60	26	73.7	167.0	2.7	–	–	[2]
9. APWP-II	60	45	74.4	191.3	2.8	57.2	–	[3]

Age is the age of the paleomagnetic or APWP pole; N is the number of independent readings of the paleomagnetic field, included into pole determination; in case of APWP poles N is the number of poles used; Lat, Long are the latitude and longitude of the paleomagnetic or APWP pole; κ and α_{95} are the precision parameter and the confidence cone of the Fisher statistics [36]; ASD is the angular standard deviation of the VGP distribution. Pole marked with asterisk (*) is transferred from Eurasia to Greenland using the pole of rotation: Lat = 52.87°, Long = 130.03°, rotation angle = −11.38° [40]. Pole marked with two asterisks (**) is transferred from North America to Greenland using the pole of rotation: Lat = 27.36°, Long = 210.59°, rotation angle = 3.57° [52]. Ages are according to geomagnetic polarity time scale of [41].

(Figs. 4 and 6b). This point is also illustrated by the different Vaigat Fm profiles having statistically distinct paleomagnetic poles with too low ASD values (Table 2, poles 1 and 2), supporting the suggestion of [17] that not even 2–300 m thick Vaigat Fm profiles represent sufficiently long periods of time to average out PSV. Secondly, it is important to identify excursions, when the geodynamo is in a weak non-dipole state and the geocentric axial dipole assumption, therefore, breaks down. It is impossible to detect that DGs 2–9 record excursional directions based solely on the 39 flows of the Svartenhuk-2 profile (Figs. 4 and 6b), and an apparently well-defined paleomagnetic pole derived from the mean of all Svartenhuk-2 flows would be ~50° off from the mean pole of this study. Thirdly, the uncertainties involved in the in-field dip/strike readings, where local effects and the lack of full three-dimensional control when observing a profile may cause dip/strike readings to be wrong. A comparison of photogrammetry and in-field dip/strike measurements (Fig. 5b) suggests that, although the in-field readings are taken at the most suitable and well-exposed locations, they are at times off by 5–6°. In-field dip/strike measurements for less well-exposed lava sequences could have errors much larger than that. Erroneous dip/strike readings are particularly problematic as they offset paleomagnetic poles without affecting the confidence limit. Accurate dip/strike photogrammetry measurements can be performed on vertical aerial photographs, which are widely available, and they can become an important asset to the paleomagnetic studies of LIPs. We note that precise photogrammetry readings not only allow us to define tectonic corrections more accurately but also show that the entire Kanisut Mb sampling area has tilted as one coherent block. Finally, it is wise to treat poles obtained prior to 1980 and the advent of detailed demagnetization experiments with particular care [8,50].

Thanks to the availability of multiple well-distributed and well-exposed flow sequences and excellent photogrammetric coverage, we have been able to take into consideration the above-described difficulties in determining our new West Greenland paleomagnetic pole. This pole is statistically indistinguishable [34] from the new reliable paleomagnetic pole of the NAIP outcropping on Faroe Islands [8] transferred to Greenland with the rotation parameters of [40] (Fig. 10a and Table 2). Moreover, PSV estimates of the two datasets are in excellent accordance (Table 2) and agree well with the global compilations of [38]. We therefore combine the West Greenland and Faroes directional data to obtain a joint-NAIP pole, which is located at Lat = 71.1°N, Long = 161.1°E in Greenland coordinates (Table 2). This pole is shown in Fig. 10b together with the two most recent updates of APWP for ~60 Ma (Table 2; [2,3]). It is important to note that these compilations include poles that are mutually inconsistent with each other and with the resulting APWP pole and therefore must be biased in a way that is not reflected in their confidence limits. The implicit assumption of the APWP compila-

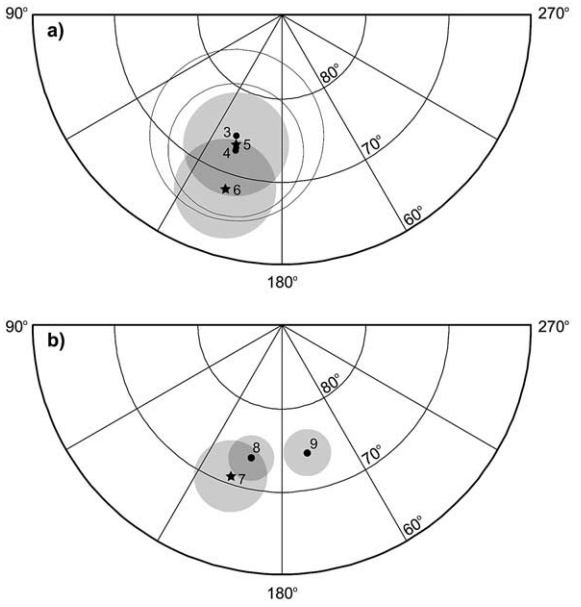


Fig. 10. (a) Equal-area projection of the Vaigat Fm and Kanisut Mb paleomagnetic poles (closed circles) together with a combined paleomagnetic pole of West Greenland and Faroe Islands paleomagnetic pole [8] rotated to Greenland (stars). (b) Equal-area projection of the joint-NAIP paleomagnetic pole (star) together with 60 Ma APWP poles of [3] and [2]. Pole numeration is as follows: 3, Vaigat Fm pole; 4, Kanisut Mb pole; 5, combined West Greenland pole; 6, Faroe Islands pole [8]; 7, joint-NAIP pole; 8, APWP pole of [2]; 9, APWP pole of [3]. For further details see Table 2.

tions that the individual biases will be averaged out when a large dataset is considered is, perhaps, dangerous unless differences are clearly caused by non-averaged PSV. This is clearly underlined by the fact that the two APWP poles are statistically different from each other. Our joint-NAIP pole is statistically indistinguishable from the ~60 Ma APWP pole of [2], which is based on 26 paleomagnetic poles from Europe and North America; but it is statistically different from the ~60 Ma APWP pole of [3], based on 45 poles from all major lithospheric plates. We suggest that the APWPs are less well defined than suggested by their confidence limits and future APWP compilations should be more restrictive.

5. Conclusions

The new paleomagnetic and photogrammetry data of the West Greenland part of the NAIP, presented in this study, demonstrate difficulties related to obtaining reliable paleomagnetic poles from LIPs. It is suggested that a substantial amount of lava flows must be sampled and accurately analyzed in order to properly average out PSV and to be able to detect excursional directions. Photogrammetry analyses of this study allowed accurate determination of dip/strike measurements of the sampled lava successions and they show that in-field dip/strike readings can have large uncertainties and cause erroneous paleomagnetic pole determination.

The Late Paleocene–Early Eocene paleomagnetic pole of West Greenland obtained in this study (Lat = 73.6°N, Long = 160.5°E; $N = 44$, $B = 383$, $\alpha_{95} = 6.2°$, $K = 13.1$) is statistically indistinguishable from the paleomagnetic pole of the Eurasian part of NAIP outcropping on Faroe Islands [8] transferred to Greenland. This good agreement reinforces the conclusions of [8] and underlines the necessity for revision of older paleomagnetic poles, many of which are found to be distinct at the 95% confidence level. These poles, used in most recent updates of APWP [2,3], could be one possible explanation for a difference between our new joint NAIP pole and some APWP poles of the same age.

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